TiberCAD Theory Module

Release 1.2.2

TiberLAB srl

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ELASTICITY

2.1 Abstract

Elasticity? is a Finite Elements solver for mechanical equilibrium problems. It brings features typically developed for structural stability solvers into device modeling. The coupled treatment of the electro-mechanical problem within a unique framework results very useful to explore for multidisciplinary ideas.

2.2 Mini theoretical intro

Assuming a small displacement, Elasticity? elasticity computed the strain by solving the equation

$$\frac{\partial}{\partial x_j} C_{ijlk} \frac{\partial u_l}{\partial x_k} = f_i$$

where C is the stiffness constant and f is the total external body force. The latter may include several contribution such as the converse piezoelectric effect the thermal expansion and a lattice mismatch induced strain. The latter, described in the example below, occurs when an interface is created between two materials with different lattice constants. Let ϵ^{LM} be the lattice mismatch strain, then the effective body force reads as :

$$f_i = -\frac{\partial}{\partial x_j} C_{ijlk} \epsilon^{LM}_{lk}$$

2.3 Example

In the following we will compute the strain induced by the lattice mismatch in a system comprising a layer of GaN between two contacts of $Al_x Ga_{1-x} N$. Once the mesh is created with gmsh (distributed along TiberCAD?) we may start to build the input file. First, we have to insert the region section

```
Device
  {
   dimension = 3
   meshfile = mesh.msh
   mesh_units = 1e-9
   material = AlGaN
   x = 0.2
   x-growth-direction = (-1,0,1,0)
   y-growth-direction = (-1,2,-1,0)
   z-growth-direction = (0,0,0,1)
```

```
   Region Well  {material = GaN}
  }
```

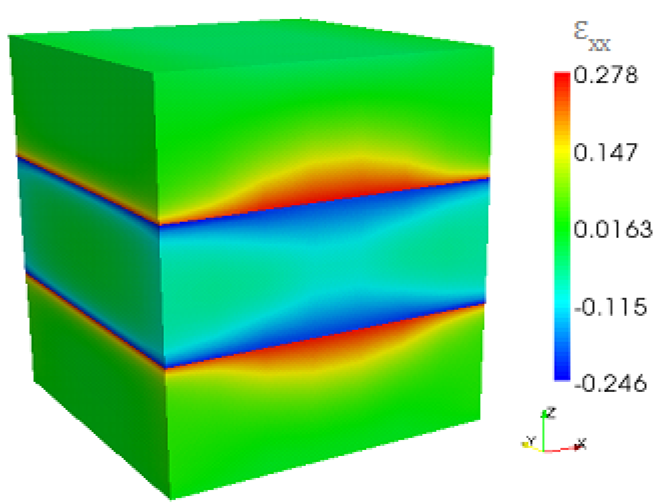
All the options included in this section, such ad the material, axis growth and alloy concentration, apply for the whole structure. If we want to specify a region with different properties, we may use the keyword Region and refer to a region name, as specified in the mesh file.

The second part of the input file is devoted to the module declaration

```
Module elasticity
  {
   Physics
     {
      BodyForcelattice_mismatch
        {
         x-growth-direction = (-1,0,1,0)
         y-growth-direction = (-1,2,-1,0)
         z-growth-direction = (0,0,0,1)
         reference_material = AlGaN
         x = 0.2
        }
      ContactClamp{regions = Base}
     }
  }
```

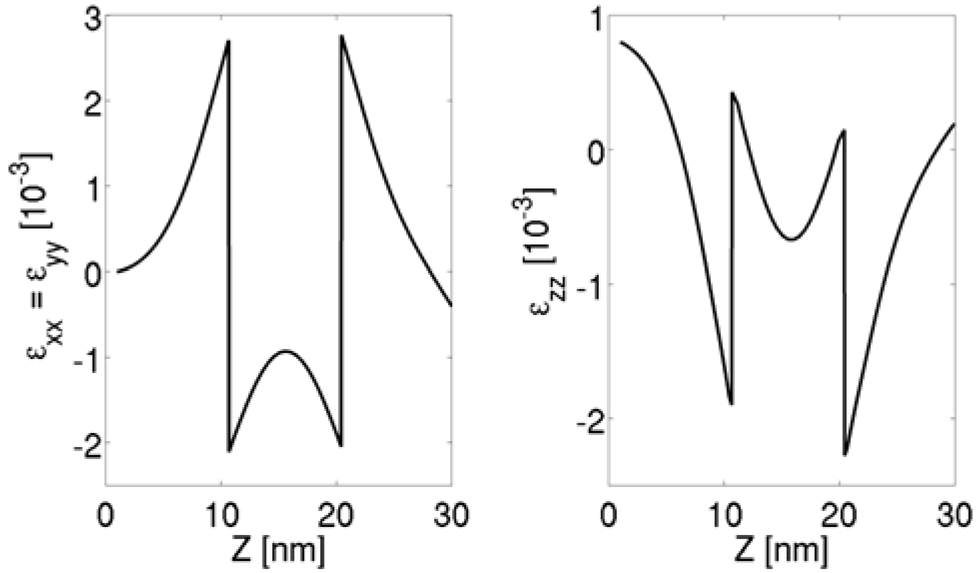
In physics we declare the physical models to be applied to our device. In our case, with BodyForce lattice mismatch we are adding a body force into the system, induced by the lattice mismatch with the reference lattice which in this case is $Al_{0.2}Ga_{0.8}N$. In order to avoid a free standing device we may want to freeze a surface. With Boundary Clamp we fix, in this case, the region Base.

The third part is simply Simulation **{solve = elasticity}** where the default name *elasticity* is used. By default, files for 3D system are written in vtu which can be read from Paraview. Output data about strain are shown.



Theoretical background

TiberCAD? computes the mechanical deformation of a body subjected to external forces by means of the equilibrium mechanical equation, i.e.



$$\frac{\partial \sigma_{ij}}{\partial x_j} = f_i$$

wheres is the stress second-rank tensor and f is the external body force applied to the system. As we will see below, any strain and stress source can be appropriately mapped into an equivalent body force.

The stress tensor e is related to the mechanical strain by means of the constitutive relationships $\sigma_{ij} = C_{ijlk}\epsilon_{lk}$ where **C** is the stiffness fourth-rank tensor. The strain is related to the displacement u by the expression

$$\epsilon_{kl} = \frac{1}{2}\left(\frac{\partial u_l}{x_k} - \frac{\partial u_k}{\partial u_l}\right)$$

Relying on the symmetry of the strain and stiffness tensor the final equation reads as

$$\frac{\partial}{\partial x_j} C_{ijlk} \frac{\partial u_l}{\partial x_k} = f_i$$

The non-linear strain is computed by applying the mechanical equilibrium equation on the deformed mesh. The number of the shape iteration is set by the keyword shape_iteration (default = 0, i.e. no deformation is performed) whereas the maximum tolerated error can be indicated with shape_error (default = 1e-3).

Stiffness constant

By default the stiffness constant is anisotropic. For the zincoblend crystal structure the independent coefficients are (with the Voigt notation) C11, C12, C44. For the wurtzite structure we have C11, C12, C13 and C44.An isotropic stiffness can be included with the keyword Stiffness isotropic. In this case, the only independent parameters are the Young modulus (Youngin GPa) and the Poisson?s ratio (Poisson).

Example:

```
Stiffness isotropic
  {
   Young = 129
   Poisson = 0.349
  }
```

Boundary conditions:

Surfaces forces f^0 are applied by imposing $\sigma_{ij}n_j = f_i^0$ along the surface with normal n. This boundary condition can be used with the keyword surface_force. The force can be specified in GPa with force. An example is shown below

```
Contact base
  {
   type = surface_force
   force = (0,0,0.5)
  }
```

On the other hand, one may want to fix some surface of the device. The keyword clamp freezes all nodes of a given surface.

Example:

```
Contactsubstrate
  {
   type = clamp
  }
```

Body forces

A constant value body force can be included by means of the keyword Body_force constant.

Example:

```
BodyForceconstant
  {
   force = (0,1,0)
  }
```

When two crystals with different crystal structure are put in contact the lattice mismatch between them may induce a strain e^LM and, therefore, a stress s=?Ce?^LM.

This stress contribution acts as a body force

$$f_i = -\frac{\partial}{\partial x_j} C_{ijlk}\epsilon_{lk}^{LM}$$

The strain source can be computed only once the reference lattice is identified. The reference material and its growth axis can be included following the same syntax of the device section.

Example:

```
BodyForcelattice_mismatch
  {
   reference_material = AlGaN
   structure = wz
   x = 0.2
   x-growth-direction = (1,0,0,0)
   y-growth-direction = (0,1,-1,0)
   z-growth-direction = (0,0,0,1)
  }
```

In presence of an electric field there might develop an additional stress source due to the so-called converse piezoelectric effect, given by :

$$\sigma_{il}^{SC} = -e_{ijk}E_k$$

The converse piezoelectric effect can be included with the keyword BodyForcepiezo and an electrostatic simulation must be indicated.

The effective body force reads as :

$$f_i = -\frac{\partial}{\partial x_j}\sigma_{ij}^{CP}$$

Example:

```
BodyForceconverse_piezo
  {
   poisson_simulation = dd
  }
```

Considering all the above mentioned force sources, the plotted total stress and strain are :

$$\sigma_{ij} = C_{ijlk}\left(\frac{\partial u_l}{\partial x_k} + \epsilon_{lk}^{LM}\right) - e_{ijk}E_k$$

$$\epsilon_{kl} = \frac{1}{2}\left(\frac{\partial u_l}{x_k} - \frac{\partial u_k}{\partial u_l}\right) + \epsilon_{lk}^{LM}$$

Reference Manual

Stiffness/Isotropic Kind: bulk Name: Stiffness Type: isotropic database_section: none Options: young (double,0.0) poisson (double [-0.5,1.0],0.0)

Stiffness/Anisotropic Kind: bulk Name: Stiffness Type: anisotropic database_section = stiffness Options: none

Body Force/constant Kind: bulk Name: BodyForce Type: constant database_section: none Options: force (double vector,(0.0,0.0,0.0))

Body Force/Lattice Mismatch Kind: bulk Name: BodyForce Type: lattice_mismatch database_section: none Options: x (double [0.0,1,0],0.0) x-growth-direction (double vector) y-growth-direction (double vector) z-growth-direction (double vector)

Body Force/Converse Kind: bulk Name: BodyForce Type: converse database_section: piezoelectricity Options: poisson_simulation (string,?none?)

Boundary/clamp Kind: interface Name: Contact Type: converse database_section: none Options: poisson_simulation (string,?none?)

Boundary/Surface Force Kind: interface Name: Contact Type: surface_force database_section: none Options: force (double vector,(0.0,0.0,0.0)

CHAPTER THREE

SIMULATION DYE SOLAR CELLS

For a brief list of literature to understand Dye Solar cells (DSC) see [Kalyanasundaram] . For a review of the model see [Gagliardi] .

The model consists in a set of drift-diflusion equations for the description of the propagation of different ions and for the electrons coupled with Poisson equation:

$$\begin{aligned}
\nabla \cdot (\mu_e n_e \nabla \phi_e) &= (G - R) && (3.1)\\
\nabla \cdot (\mu_{I^-} n_{I^-} \nabla \phi_{I^-}) &= -\frac{3}{2}(G - R) && (3.2)\\
\nabla \cdot (\mu_{I_3^-} n_{I_3^-} \nabla \phi_{I_3^-}) &= \frac{1}{2}(G - R) && (3.3)\\
\nabla \cdot (\mu_c n_c \nabla \phi_c) &= 0, && (3.4)
\end{aligned}$$

where μ_α refers to carrier mobilities, n_α to concentrations and ϕ_α to electrochemical potentials. *R* is the recombination term and *G* the generation term due to illumination. The Poisson equation to handle the internal potential drop:

$$-\varepsilon \triangle \varphi = q[n_c + N_D^+ - n_{I^-} - n_{I_3^-} - (n_e - \bar{n}_e)], \tag{3.5}$$

where N_D^+ is the amount of ionized dyes and it is equal to:

$$N_D^+ = \frac{G}{k_3} \tag{3.6}$$

with *G* the generation term and :math‘k_3‘ the rate constant of dye regeneration. The dielectric constant, ϵ , of the mesoporous material is a mix the two dielectric functions of the semi- conductor and the electrolyte. We use the Maxwell-Garnet model where the dielectric function of the mixed medium becomes:

$$\varepsilon = \varepsilon_s \frac{\varepsilon_e + 2\varepsilon_s + 2\epsilon_p \varepsilon_e - 2\varepsilon_s \epsilon_p}{\varepsilon_e + 2\varepsilon_s - \epsilon_p \varepsilon_e + \varepsilon_s \epsilon_p} \tag{3.7}$$

where ϵ_s and ϵ_e are the dielectric constants of the semiconductor and the electrolyte, respectively, and ϵ_p is the porosity of the medium. The recombination term depends largely on the loss mechanisms at the electrolyte/oxide interface which follows the reac- tion chain:

$$\begin{aligned}
I^- &\rightleftharpoons I + e && (3.8)\\
2I &\rightleftharpoons I_2 && (3.9)\\
I_2 + I^- &\rightleftharpoons I_3^-. && (3.10)
\end{aligned}$$

From the chemical path we can get a formula for the interface recombination (considering that the flrst chemical reaction is the slow process):

$$R = k_e \left[\left(\frac{n_e}{\bar{n}_e} \right)^{\beta} \bar{n}_e \sqrt{\frac{n_{I_3^-}}{n_{I^-}}} - \bar{n}_e \sqrt{\frac{\bar{n}_{I_3^-}}{\bar{n}_{I^-}^3} n_{I^-}} \right], \tag{3.11}$$

where the electron rate k_e is the recombination rate constant.

For the boundary conditions of the model we assume at the photoanode: * $\phi_e = V$: electrochemical potential of electrons set with the voltage applied; * $\nabla \phi_{I^-} = 0$: no iodide current at the photoande; * $\nabla \phi_{I_3^-} = 0$: no triiodide current at the photoande; * $\nabla \phi_c = 0$: no cationic current; * $\nabla \varphi = 0$: no charged layer at the photoanode;

at the cathode:

- $\nabla \phi_e = 0$: no electronic current at the cathode;
- $-q\mu_{I^-} n_{I^-} \nabla \phi_{I^-} = \frac{3}{2} \left(\frac{-E_{red}(\mathbf{r_c})}{R_L} \right)$: split of the current between the ionic species;
- $-q\mu_{I_3^-} n_{I_3^-} \nabla \phi_{I_3^-} = -\frac{1}{2} \left(\frac{-E_{red}(\mathbf{r}_c)}{R_L} \right)$: split of the current between the ionic species;
- $\nabla \phi_c = 0$: no cationic current;

integral boundary conditions for conservation of ionic species:

- $\int_\Omega \left[\frac{1}{3} n_{I^-}(\mathbf{r}) + n_{I_3^-}(\mathbf{r}) \right] d\mathbf{r} = \left(\frac{1}{3} \bar{n}_{I^-} + \bar{n}_{I_3^-} \right) \Omega$: conservation of iodine ions within the cell;
- $\int_\Omega n_c(\mathbf{r}) d\mathbf{r} = \bar{n}_c \Omega$: conservation of cation within the cell;

where ω is the volume of the cell, $n\alpha$ the density of charged species and the index α stands for cation (c), iodide (I^-), triiodide (I_3^-) and electrons (e). R_L is the external load. The bias applied is equal to:

$$V = \phi_e - E_{red}. \tag{3.12}$$

E_{red} is the redox potential. The redox potential can be evaluated using a Nernst approx- imation:

$$V = \phi_e - E_{red}. \tag{3.13}$$

BIBLIOGRAPHY

[Kalyanasundaram] Kalyanasundaram Kuppuswamy, “Dye-Sensitized Solar Cells”, *Editor Crc Pr Inc.* , 2010.

[Gagliardi] Alessio Gagliardi, Matthias Auf der Maur, Desiree Gentilini and Aldo Di Carlo, “Modeling of Dye sensitized solar cells using a finite element method”, *J. Computational Electronics* , vol. 8, pp. 12, 2009.